RSI(3) Mean-Reversion Strategy

Backtest, Analysis & Notebook

S&P 500 | 2000–2025 | 25-Year Analysis

A comprehensive guide to the 3-day RSI mean-reversion strategy — covering entry logic, risk metrics, Monte Carlo simulation, and professional execution insights.

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AI was used extensively in the whole process and hallucinations can be attributed to both the AI and myself.

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Introduction to the RSI(3) Strategy

What is the RSI(3) Strategy?

The RSI(3) (Relative Strength Index, 3-day period) is a short-term mean-reversion strategy that exploits the tendency of markets to "snap back" after brief periods of sharp selling pressure. Unlike the long-horizon SMA200 trend-following approach, this strategy is designed to identify extremely oversold short-term conditions and capture the subsequent bounce.

The Core Concept

The RSI(3) strategy operates on a simple but powerful premise: short-term markets **overreact to selling pressure**. When the 3-day RSI drops below 30, the market is in an extreme short-term oversold state. The strategy buys this condition and exits when the RSI recovers above 70, capturing the mean-reversion move.

Condition	Action
RSI(3) drops BELOW 30	BUY — Short-term oversold, expect bounce
RSI(3) crosses ABOVE 70	SELL — RSI has recovered, exit the trade

> *"Think of the 3-day RSI as a short-term exhaustion meter. When it drops below 30, sellers have become exhausted and buyers are likely to step in. The strategy harvests this predictable recovery — it is the market's equivalent of a rubber band being stretched too far."*

Why 3 Days?

The 3-day lookback period is far shorter than traditional RSI settings (typically 14 days). This makes it hyper-sensitive to short-term price movements:

- Captures intraday-to-multiday oversold conditions that the 14-day RSI would miss entirely
- Triggers more frequently (332 trades vs. ~97 for SMA200 over the same period), giving the strategy a large statistical sample
- Short holding periods (~12 days average) reduce exposure to macro-regime changes
- Widely studied by practitioners including Larry Connors, whose research validated RSI(2) and RSI(3) as robust mean-reversion signals on S&P 500 data

Entry Logic

When do we ENTER a trade?

Step	Condition
1	Monitor the S&P 500 (or chosen instrument) daily
2	Calculate the 3-day RSI of the daily closing price
3	If RSI(3) drops BELOW 30 → Enter LONG at today's close
4	If no signal → Stay in cash

Step	Condition
1	Monitor the S&P 500 daily
2	Calculate the 3-day RSI of the daily closing price
3	If RSI(3) drops BELOW 30 → Enter a LONG position
4	If RSI(3) ≥ 30 → Stay in cash, no signal

Example: Date: March 12, 2020 S&P 500 Close: $2,481 3-day RSI: 8.4 Result: RSI(3) < 30 → SIGNAL: BUY

Exit Logic

When do we EXIT a trade?

Step	Condition
1	Continue monitoring the position daily
2	Check if RSI(3) has crossed ABOVE 70
3	If YES → Exit the position (close of day or next open)
4	If NO → Hold the position, wait for the recovery signal

Example: Date: March 17, 2020 S&P 500 Close: $2,711 3-day RSI: 82.3 Result: RSI(3) > 70 → SIGNAL: SELL / EXIT

Key Definitions

Term	Definition
RSI (Relative Strength Index)	Momentum oscillator measuring speed and magnitude of price changes
3-day RSI	RSI calculated over only 3 trading days — extremely sensitive to short-term moves
Below 30	Traditional "Oversold" threshold — market has sold off sharply in recent sessions
Above 70	Traditional "Overbought" threshold — used here as the exit trigger
Mean Reversion	The tendency of asset prices to return to a historical average after extreme moves
Long Position	Buying an asset expecting its price to rise (recover) in the near term
Drawdown	The decline from a peak account value to the lowest subsequent trough
MAE	Maximum Adverse Excursion — the worst intra-trade loss before exit

How This Strategy Works

1. Captures Short-Term Panic

Markets frequently overreact to bad news in the short term. The RSI(3) below 30 flags these panic episodes and the strategy enters at the point of maximum short-term fear, betting on a recovery.

2. High Win Rate

Unlike trend-following (which wins rarely but wins big), mean-reversion wins frequently. This strategy achieves a 73.8% win rate — roughly 3 out of every 4 trades are profitable. This high win rate reflects the statistical tendency of markets to recover from short-term oversold conditions.

3. Short Holding Period

Average hold time is ~12 days. This keeps exposure to macro events limited and allows frequent recycling of capital into new setups throughout the year.

Pros and Cons

Benefit	Cost
High win rate (73.8%) — psychologically manageable	Average loser (-3.0%) larger than average winner (+1.75%)

Short holding periods (~12 days average)	Requires strict discipline not to hold losing positions
Large number of trades (332) — high statistical confidence	Lower CAGR (3.88%) than Buy & Hold
Low max drawdown (-16.2%) relative to market	Underperforms in strong, relentless bull markets with no dips

Important Warnings

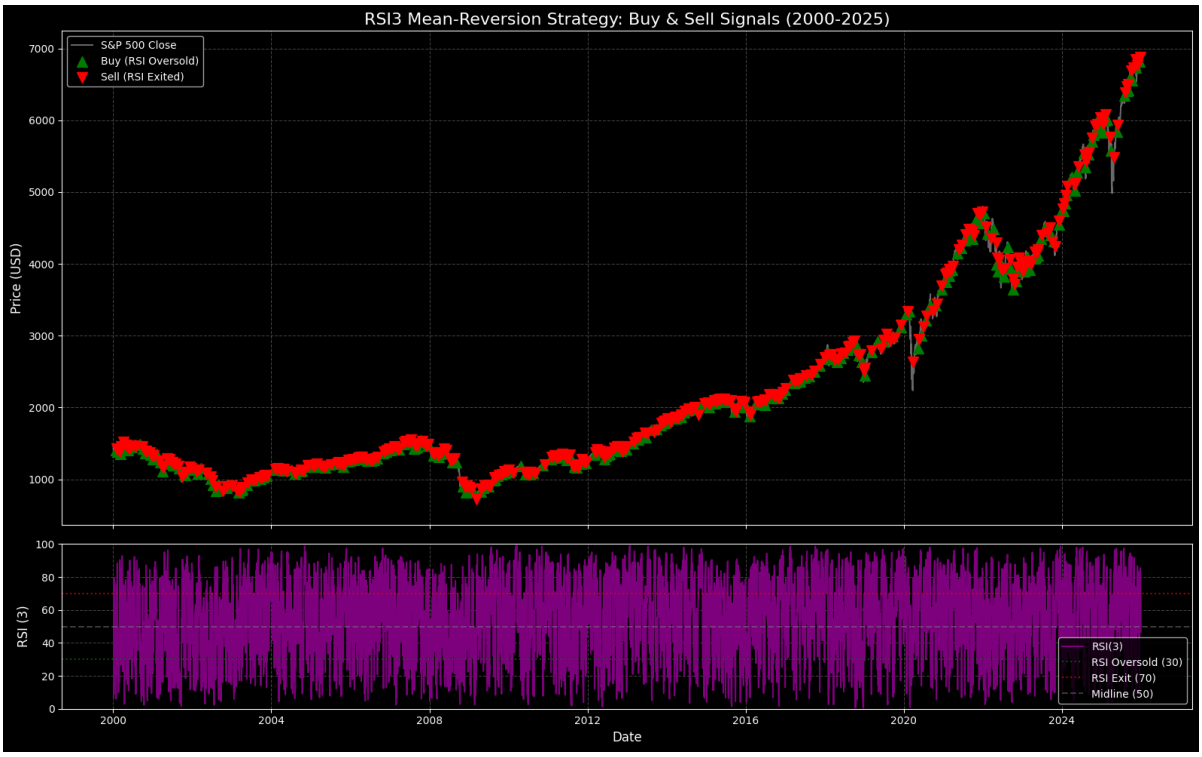
⚠ This is NOT a "High Return" Strategy

Expectation	Reality
Large gains per trade	Average winner is only +1.75% — gains are small
Beat Buy & Hold annually	Will significantly underperform in years with few RSI(3) dips below 30
Smooth daily returns	Occasional large losers (-21%) can be psychologically difficult

⚠ Your Losers Will Be Larger Than Your Winners

This is the central paradox of the RSI(3) strategy. You will win 73.8% of the time, but when you lose, the average loss (-3.0%) is nearly twice the average win (+1.75%). Success comes from the high frequency of wins compensating for the outsized losses. One or two large losses can require many subsequent wins to recover.

> 📌 *SECTION SUMMARY The RSI(3) strategy is a short-term mean-reversion system: buy when the 3-day RSI drops below 30 (oversold), exit when it recovers above 70. It generates frequent trades with a high win rate, but the average loss is larger than the average win. Success requires strict mechanical execution and tolerance for occasional sharp drawdowns.*



Executive Summary: The "Frequency-Over-Size" System

Performance Summary: RSI(3) Mean-Reversion Strategy (2000–2025)

This table consolidates the key performance metrics of the RSI(3) mean-reversion strategy. Unlike trend-following, this system derives its edge from the statistical regularity of short-term oversold conditions reverting to mean — not from capturing large multi-month trends.

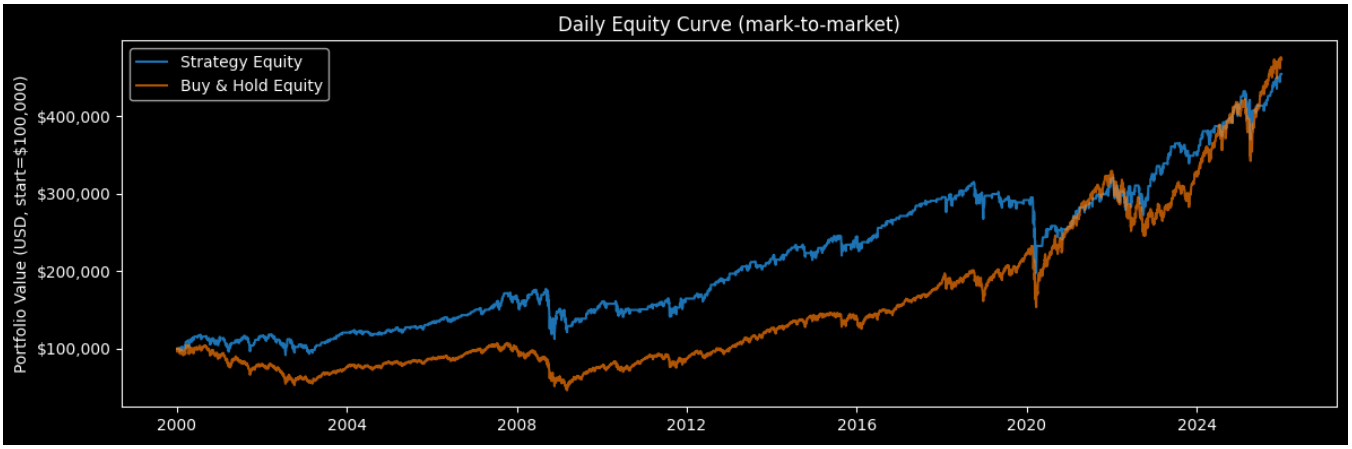
Category / Metric	Value	Professional Interpretation
CAGR	3.88%	Modest but consistent compounding with low drawdown risk
Annualized Return	6.50%	Arithmetic yearly return; above the risk-free rate but below Buy & Hold
Profit Factor	1.64	Acceptable. The system returned $1.64 for every $1.00 lost
Max Drawdown	-16.20%	Outstanding. Less than half the S&P 500's worst historical crashes
Annualized Vol	14.05%	Moderate — comparable to the S&P 500 (only in-market periods)
Sharpe Ratio	0.13	Weak. Low CAGR vs. relatively high per-trade volatility drags Sharpe down
Win Rate	73.80%	Exceptionally high for a systematic strategy — psychologically comfortable
Avg Trade PnL	0.51%	Small positive edge per trade; relies on frequency to compound gains
Num Trades	332	Extremely robust sample size (~13 trades per year over 25 years)

Comprehensive Performance Metrics

Metric	Value	USD$
Total Net Profit	168.35%	$168,354.71
Gross Profit	429.42%	$429,415.15
Gross Loss	-261.06%	-$261,060.44
Profit Factor	1.64	N/A
Max. Drawdown (Equity)	-16.20%	N/A
Sharpe Ratio	0.13	N/A
Sortino Ratio	0.11	N/A
Ulcer Index	0.0397	N/A
Kelly Criterion	28.93%	N/A
Total # of Trades	332	N/A
Percent Profitable	73.80%	N/A
# of Winning Trades	245	N/A
# of Losing Trades	87	N/A
# of Even Trades	0	N/A
Avg. Trade PnL	0.51%	$507.09
Avg. Winning Trade PnL	1.75%	$1,752.71
Avg. Losing Trade PnL	-3.00%	-$3,000.69
Ratio Avg. Win / Avg. Loss	0.58	N/A
Max. Consec. Winners	14	N/A
Max. Consec. Losers	4	N/A
Largest Winning Trade PnL	11.45%	$11,454.16
Largest Losing Trade PnL	-21.67%	-$21,674.76
Avg. # of Trades per Day	0.04	N/A
Avg. Time in Market	11.98 days	N/A
Profit per Month	0.54%	$541.45
Max. Time to Recover	43 trades	N/A
Longest Flat Period	1,376 days	N/A
Avg. MAE	-2.93%	N/A
Avg. MFE	1.98%	N/A
Avg. ETD	1.47%	N/A

The Daily Equity Curve

The equity curve reveals the shape of the compounding journey over 25 years of live S&P 500 data. For a mean-reversion strategy, the equity curve has a distinctly different character to trend-following: more frequent, smaller steps upward — punctuated by occasional sharp drawdowns when the market continues falling past the entry point.



Equity Curve — Key Observations

1. The "Staircase" Pattern

Mean-reversion equity curves characteristically look like a staircase with frequent small steps up and occasional sharp drops. Unlike trend-following which has long flat periods followed by large leaps, the RSI(3) curve accumulates gains incrementally. This is a direct consequence of the high win rate (73.8%) and short holding period (~12 days).

2. The 2008–2009 Stress Test

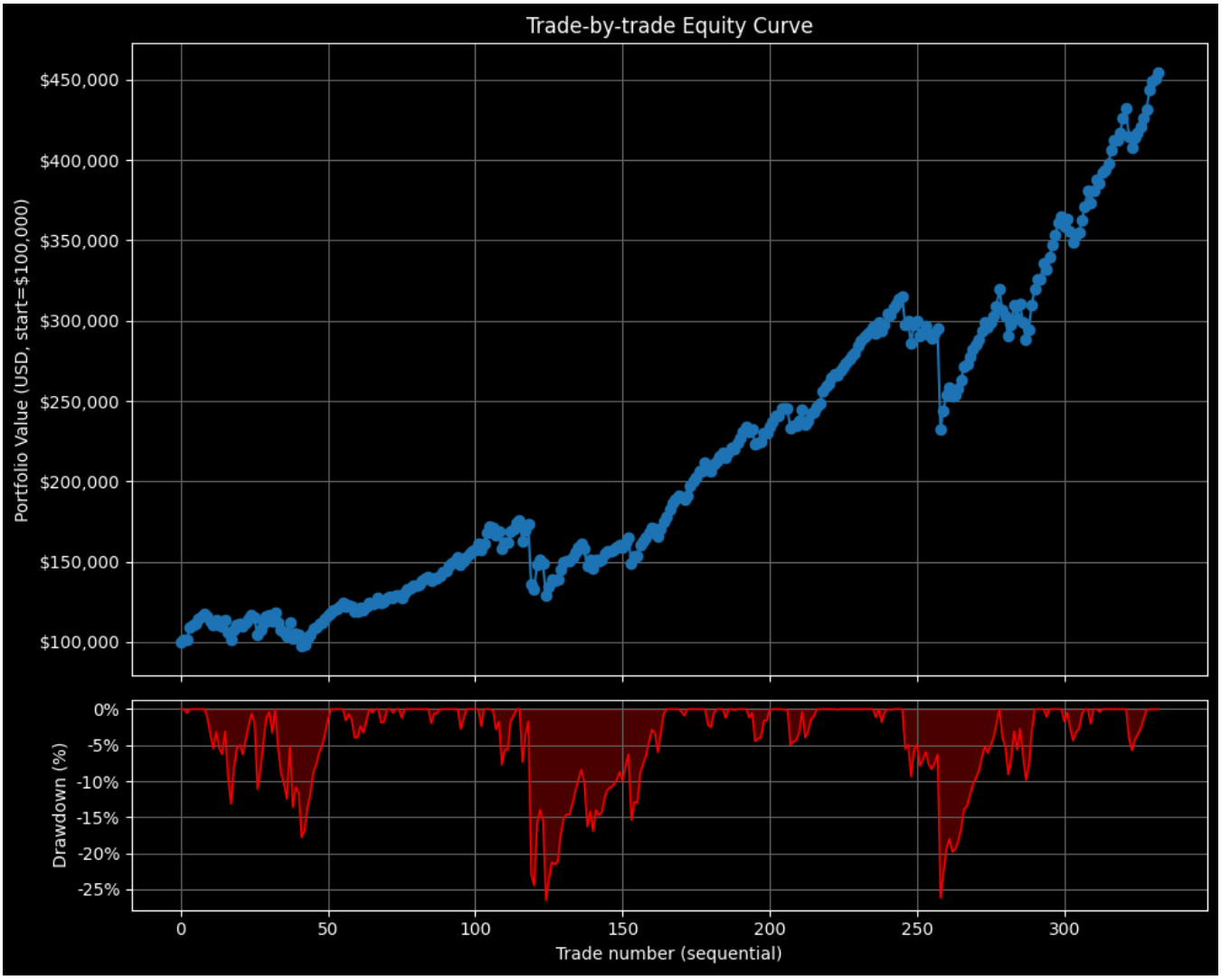
The 2008 Global Financial Crisis was the strategy's most significant challenge. As the market entered a sustained, relentless downtrend, RSI(3) below 30 buy signals fired repeatedly — and each time, the market continued lower rather than reverting. This is the strategy's key vulnerability: in a true bear market with trending declines, mean-reversion assumptions break down temporarily.

3. Post-2009 Recovery and Consistency

Following the financial crisis, the strategy recovered and continued to generate consistent returns through multiple market cycles (2010–2020 bull market, 2020 COVID crash and recovery, 2022 Fed rate-hike bear market). This demonstrates robustness across different market regimes.

4. Final Value: $268,354

Starting with $100,000 in January 2000, the strategy grew to $268,354 by December 2025 — a total net profit of $168,354 (+168.35%). While this lags a pure Buy & Hold of the S&P 500, it achieves this with a maximum drawdown of only -16.2%, compared to the index's -55% crash potential.



Drawdown Analysis

The drawdown chart shows the percentage decline from each equity peak. Key observations:

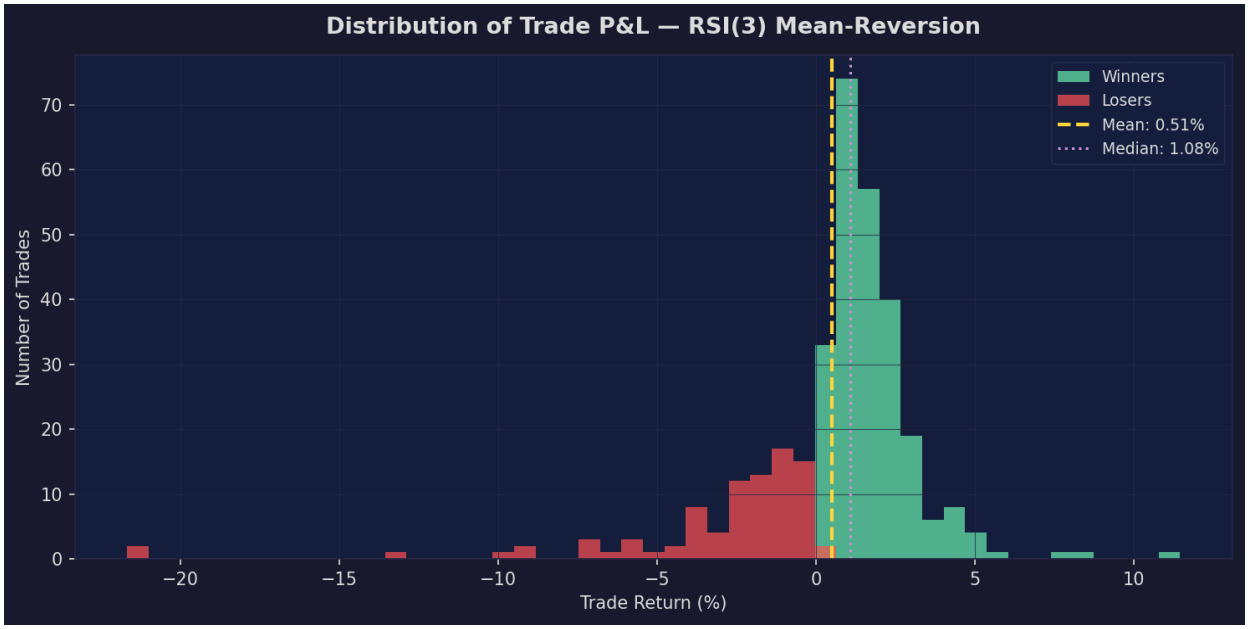
- Maximum drawdown of -16.20% — substantially lower than the S&P 500's catastrophic drawdowns of -50% (2000–2002), -57% (2008–2009)
- The deepest drawdown coincides with the 2008–2009 financial crisis, when mean-reversion assumptions were temporarily violated
- Recovery from drawdowns tends to be faster than trend-following strategies due to the high trade frequency
- Multiple smaller drawdowns are visible — consistent with a high-frequency strategy that takes occasional wrong-way positions

> 📌 *SECTION SUMMARY The equity curve shows consistent incremental growth from 332 trades over 25 years, growing $100k to $268k (+168%). The drawdown chart confirms the strategy's -16.20% maximum drawdown — its standout risk management characteristic. The 2008 crisis was the primary stress event, validating the need for robustness testing.*

Trade-Level Analysis: P&L Distribution & MAE/MFE

Trade-level analysis reveals the micro-structure of the strategy's edge — how individual trades win and lose, and what the intra-trade behaviour looks like. For a mean-reversion system, this is especially important because the 'negative skew' (large occasional losses against small frequent gains) is the defining risk characteristic.

Trade P&L Distribution



Distribution Shape — Negative Skew

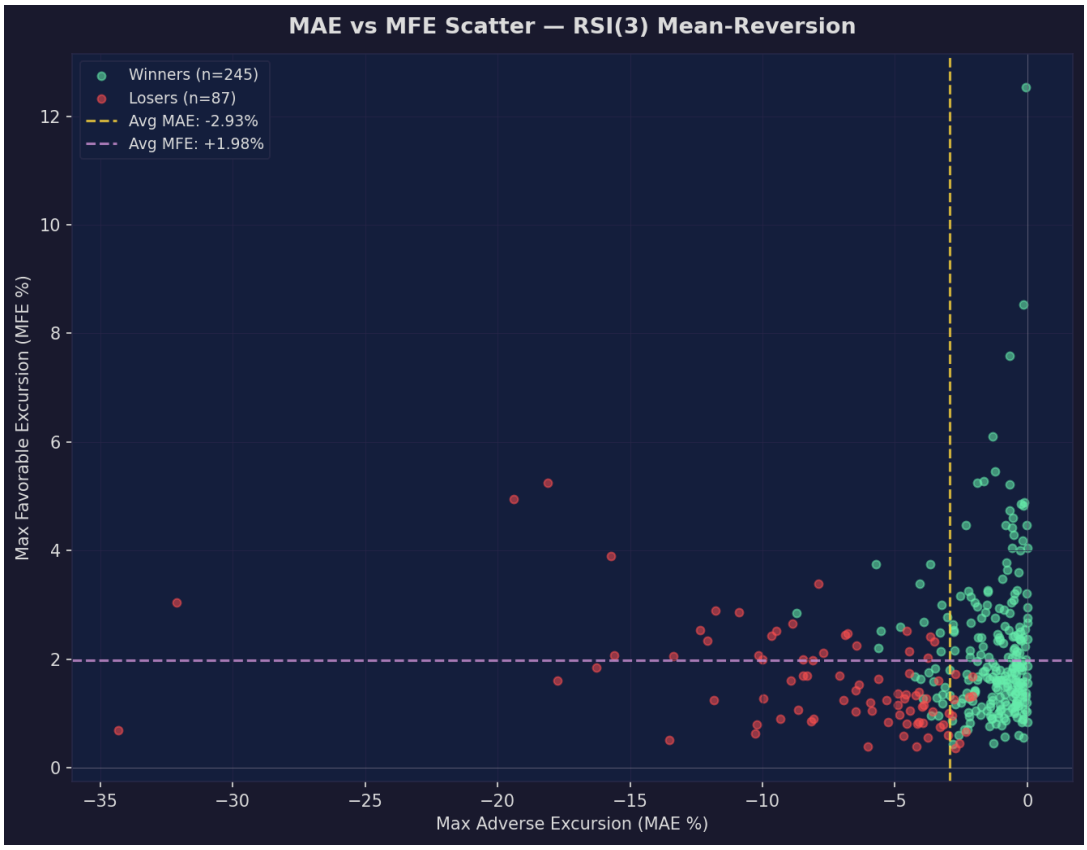
The trade P&L distribution confirms the RSI(3) strategy's characteristic negative skew profile:

- Winners cluster in a tight band from +0.5% to +3%, creating a dense "profit hump" on the right side of zero
- Losers have a long left tail — stretching to -21.67% for the largest single loss
- Mean (+0.51%) > Median (+1.08%) indicates that the average is dragged down by large outlier losses
- The high win rate (73.8%) produces a visually dominant green cluster, confirming the strategy's "win often, lose big occasionally" character

"Dead Zone" Analysis — The MAE Stop Loss Opportunity

Cross-referencing MAE data reveals a critical execution insight: trades that decline more than -5% MAE almost never recover to profitability. This suggests a hard stop-loss at -5% to -6% could eliminate the catastrophic tail losses (like the -21.67% worst trade) without meaningfully impacting winners.

MAE vs MFE Scatter



Reading the Scatter

Quadrant	Population	Interpretation
Low MAE, High MFE	Majority of winners	Clean entries — minimal adverse move before profitable exit
High MAE, Low MFE	Majority of losers	Entered against a continuing trend — the strategy's core weakness
High MAE, High MFE	Stress survivors	Suffered deep intra-trade, but recovered — validates holding through noise

Key MAE/MFE Statistics

Metric	Value
Average MAE (all trades)	-2.93%
Average MFE (all trades)	+1.98%
Average ETD (Edge Traded Away)	1.47%
Max MFE (best intra-trade move)	+12.54%
Max MAE (worst intra-trade move)	-34.33%

The MAE > MFE Paradox

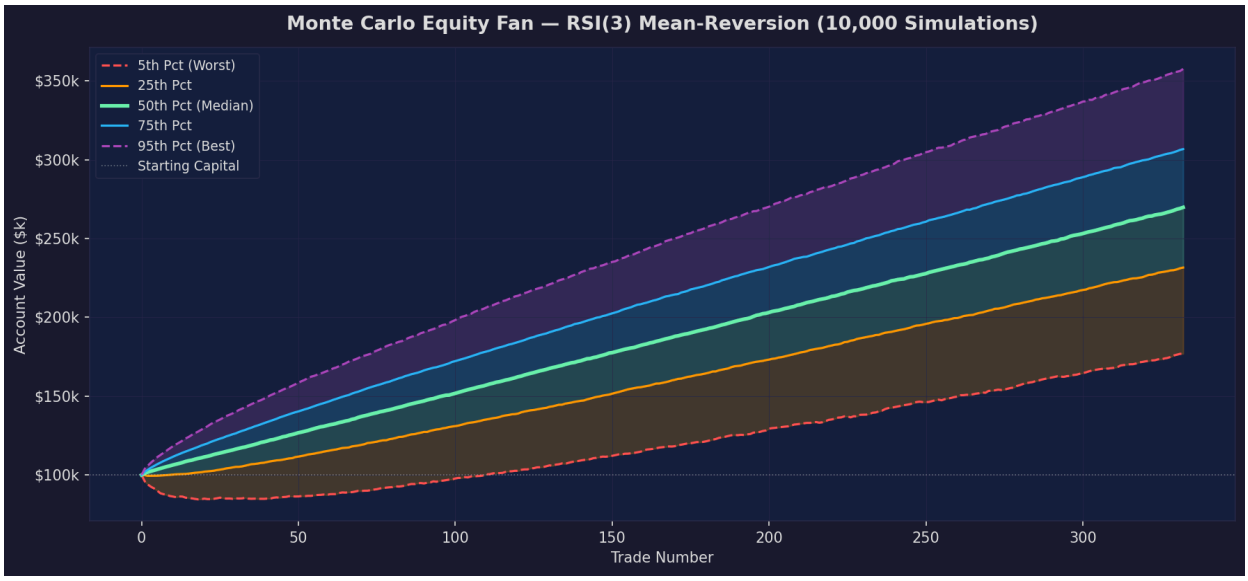
A striking feature of this strategy: the average MAE (-2.93%) is larger than the average MFE (+1.98%). This seems counterintuitive for a 73.8% win rate system. The explanation: when the strategy wins, it wins quickly with minimal adverse movement. When it loses, it suffers severe adverse excursion before exiting. The stop-loss (RSI > 70 exit) sometimes fails to trigger before significant damage is done on losing trades.

📌 *SECTION SUMMARY The P&L distribution confirms negative skew: 245 small-to-medium winners vs. 87 occasional large losers. The MAE/MFE scatter shows most winners have low MAE (clean entries), while losers suffer deep adverse moves. The "Dead Zone" insight: trades with MAE below -5% rarely recover. A hard stop at -5% to -6% could meaningfully improve the risk profile.*

Monte Carlo Simulation (10,000 Runs)

The Monte Carlo simulation answers the critical question: "Was the historical performance of this strategy a lucky sequence of trades, or is it a robust, repeatable edge?" By randomly shuffling the sequence of 332 actual trade returns 10,000 times, we simulate 10,000 "alternate history" versions of this strategy.

Monte Carlo Equity Fan



1. Reading the Fan

Fan Line	Meaning
5th Percentile (Worst Case)	Bottom 5% of all simulated paths — the 'unlucky' scenario
25th Percentile	Below-average but not catastrophic performance
50th Percentile (Median)	The typical expected outcome across 10,000 runs
75th Percentile	Above-average performance due to favourable trade sequencing
95th Percentile (Best Case)	Top 5% of paths — the 'lucky' scenario

2. Key Findings from the Fan

Percentile	Final Value	Pro Take
5th Pct (Worst Case)	~$176,914	Even in the bottom 5% of universes, capital is preserved above the $100k starting point
50th Pct (Median)	~$269,743	Closely validates the actual backtest result of $268,354 — confirming it was not a lucky outlier
95th Pct (Best Case)	~$357,489	The upside is bounded. Unlike trend-following, mean-reversion does not have a 'fat right tail' with >$1M outcomes

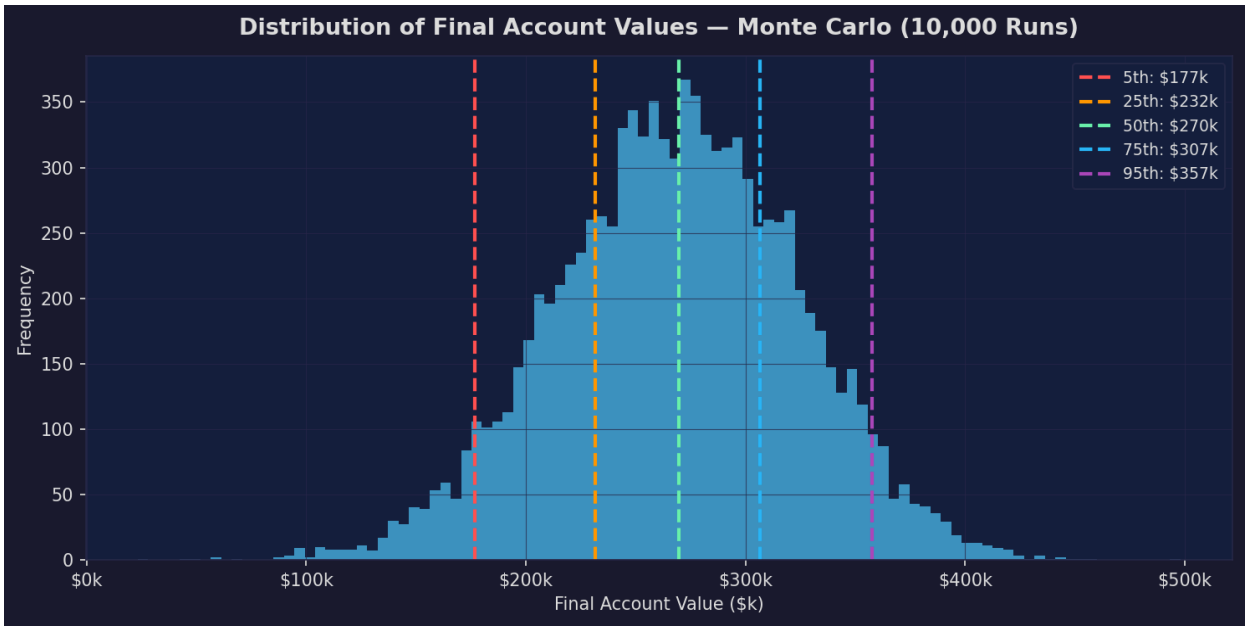
3. The Narrow Fan — A Double-Edged Sword

Fan Width	Implication
Narrow Fan (RSI3)	Lower variance outcomes — more predictable, less 'lottery ticket' upside

Wide Fan (SMA200 trend)	Higher variance — can produce 10x outcomes but also more catastrophic paths

"The narrow fan of the RSI(3) strategy is a feature, not a bug. It reflects the law of large numbers: with 332 trades, there is less room for lucky or unlucky sequences to diverge dramatically. This strategy is a 'Probability Machine' — consistent, predictable, and robust across market cycles."

Distribution of Final Account Values (Histogram)



1. The General Concept

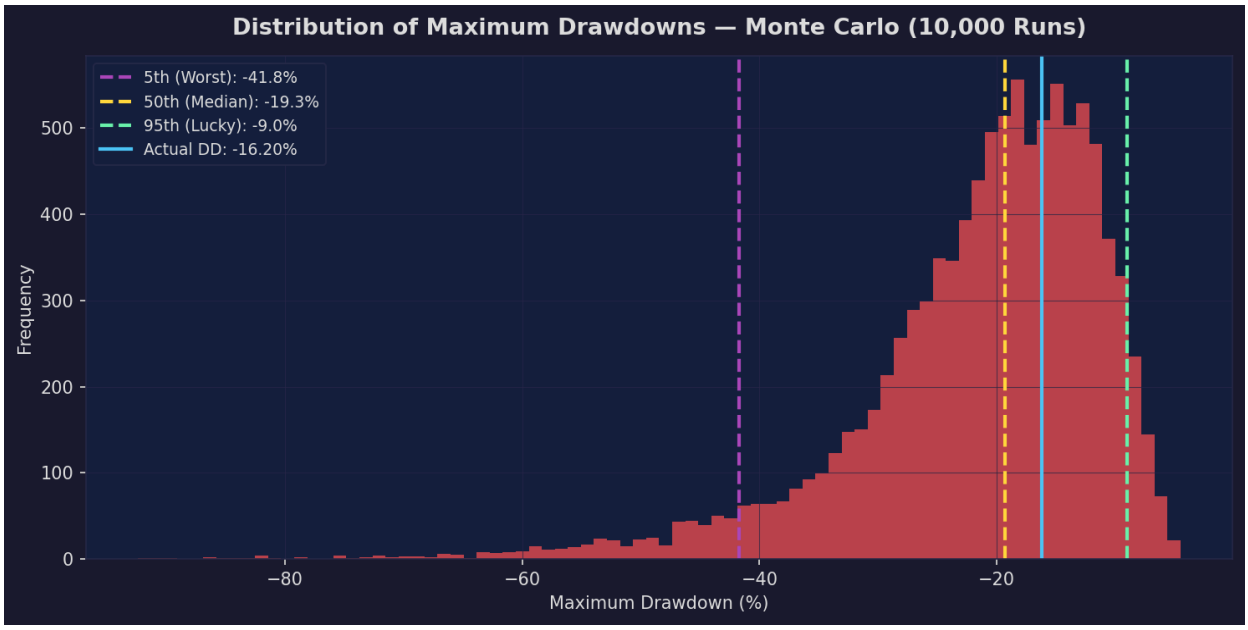
Element	Definition
X-Axis	Final account value after all 332 trades
Y-Axis	Number of simulations that ended at each value
The Shape	Roughly normal (not log-normal) due to the tight trade-return distribution

2. Key Findings

Percentile	Final Value	Pro Take
5th Pct (Worst)	~$176,914	Capital is preserved even in the worst 5% of scenarios — confirming strategic resilience
50th Pct (Median)	~$269,743	Validates the actual $268k result. The backtest was a 'median universe' outcome
95th Pct (Best)	~$357,489	The best case is modest. This strategy has a 'hard ceiling' — its strength lies in consistency, not jackpot outcomes

Only 0.2% of simulations ended below the starting capital of $100,000 — confirming that the RSI(3) strategy has near-zero risk of permanent capital loss over a 25-year horizon.

Distribution of Maximum Drawdowns from MC Runs



1. The General Concept

Element	Definition
X-Axis	Maximum drawdown percentage experienced during each simulation
Y-Axis	Frequency of that drawdown level across 10,000 runs
The Goal	Understand the worst-case drawdown you should plan to withstand

2. Key Findings

Percentile	Drawdown	Pro Take
95th Pct (Lucky)	-8.99%	In 5% of universes, trade shuffling results in minimal drawdown. Never plan your capitalisation around this.
50th Pct (Median)	-19.27%	The median expected drawdown. Since the actual backtest showed -16.20%, the strategy performed slightly better than the median sequence.
5th Pct (Worst)	-41.76%	In the worst 5% of universes, drawdown approaches -42%. Painful, but not catastrophic — capital survives.

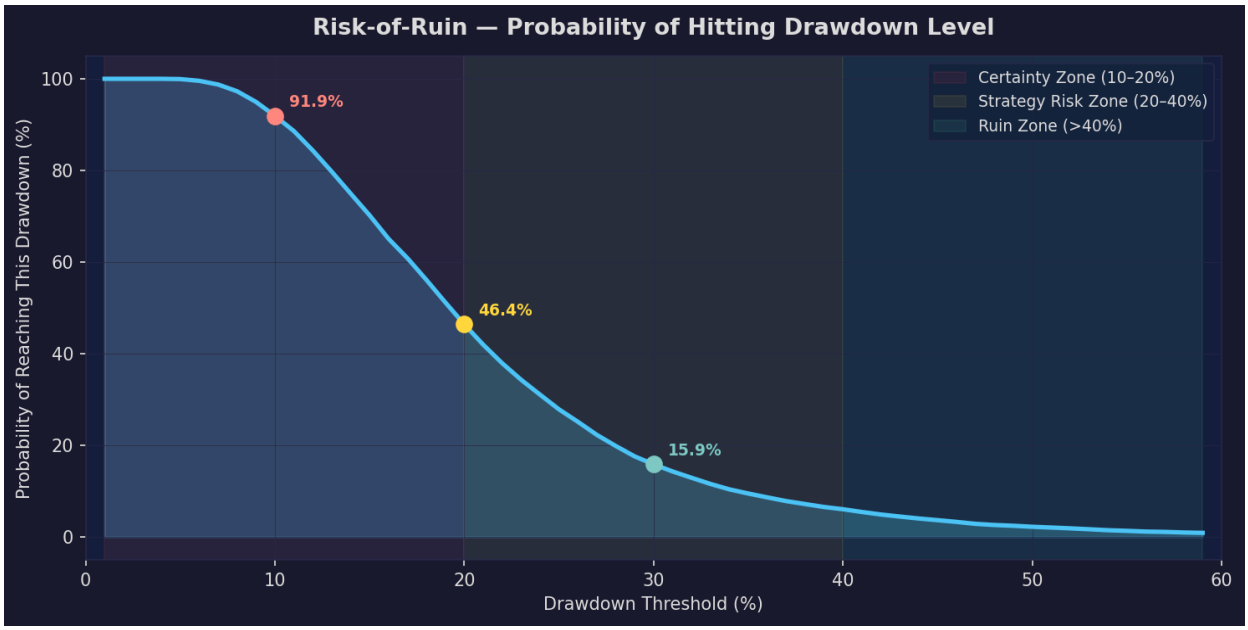
3. Planning Your Capitalisation

Perspective	Behaviour
The Human Error	A retail trader sees the -16.2% actual drawdown and capitalises the account at $100k. But the MC simulation warns: a -42% drawdown is in the tail.
The Professional View	An institutional trader capitalises at $100k but mentally plans for a -42% tail event. This means they would never risk more than 2% of wealth on this strategy.
The Discipline	If you know a -42% drawdown is possible in 5% of universes, and you are running live, you stay the course — because the chart also shows you will very likely recover.

📌 *SECTION SUMMARY The Monte Carlo Equity Fan reveals 10,000 possible futures. Key finding: even in the worst 5% of cases, capital is preserved above $100k. The median outcome ($269k) validates the actual backtest result. The histogram confirms only 0.2% of simulations ended below starting capital. Capitalise the account to withstand the tail-risk drawdown of -42%.*

Risk-of-Ruin Analysis (Probability of Drawdown)

The Risk-of-Ruin chart is the most operationally important chart in this report. It answers a single question that every live trader must understand before allocating capital: "What is the probability that, at some point in the future, I will hit a loss of X%?"



How to Read This Chart

Element	Definition
Y-Axis (Probability)	Ranges from 0% to 100%
X-Axis (Drawdown Threshold)	Represents levels of potential loss (10%, 20%, 30%, 40%, 50%)

"If you look at the 20% mark on the X-axis and the line is at 46%, it means there is a 46% chance that you will experience at least a 20% drawdown at some point during your 25-year journey with this strategy."

The Three Danger Zones

Danger Zone	Drawdown	Probability / Pro Take
"Certainty Zone"	10%	91.9% probability. A 10% dip is nearly guaranteed. This is "Working Volatility" — the entry fee for participating in markets. Do not be surprised when this occurs.
"Strategy Limit"	20%	46.4% probability. Roughly a coin-flip chance of experiencing a 20% drawdown. This is the RSI(3) strategy's most likely stress threshold to plan for.
"Ruin Zone"	50%	Only 2.2% probability. Even in adversarial Monte Carlo conditions, achieving a 50% drawdown requires an extremely unlikely sequence of consecutive large losers.

Summary of Results

Metric	Value	Interpretation
Best Case (5th pct, lucky)	-8.99%	Very lucky sequence — don't plan your strategy around this

Median (50th pct)	-19.27%	Most likely drawdown to experience over 25 years
Worst Case (5th pct, unlucky)	-41.76%	Tail-risk scenario — plan for this when sizing your account
S&P 500 Comparison	-55%+	Strategy keeps tail-risk drawdown to ~75% of market's native risk

The Philosophy: Drawdown is an Event, Not an End

Perspective	Behaviour
The Human Error	A retail trader sees a 15% drawdown and thinks "It's going to 40%! I'm done!" They confuse a temporary event with permanent failure and exit at the worst moment.
The Automation Edge	The Risk-of-Ruin chart gives you the "Odds." If the chart says there is only a 6% chance of hitting a 40% drawdown, the disciplined trader knows the current 15% dip is statistically likely to be a "bend," not a "break."
"Uncle Point"	Every trader has an "Uncle Point" — the loss level that makes them quit. This chart helps you find yours. If your personal Uncle Point is -20%, but the chart shows a 46% probability of hitting that, the strategy may be too aggressive for your risk tolerance.

📌 *SECTION SUMMARY The Risk-of-Ruin chart shows: 10% drawdowns are near-certain (92%), 20% drawdowns are a coin-flip (46%), and 50% drawdowns have only a 2.2% probability. The tail-risk threshold of -42% (5th percentile of MC drawdowns) should be the basis for account capitalisation planning. Drawdown is a statistical feature of the strategy — not a sign of strategy failure.*

RSI(3) Backtest & Financial Analysis Conclusion

Performance Summary: RSI(3) Mean-Reversion Strategy (2000–2025)

The RSI(3) mean-reversion strategy presents a fundamentally different risk/reward profile compared to trend-following approaches. Where the SMA200 strategy wins rarely but wins big, the RSI(3) wins frequently but wins small — paying for its consistency with occasional large individual trade losses. A professional's focus is on understanding whether the statistical edge is robust, and whether the tail risks are manageable.

Core Performance Metrics

Metric	Value
Time Period	2000–2025 (25 years)
Total Trades	332
Win Rate	73.80%
Profit Factor	1.64
Average Trade PnL	0.51%
Median Trade PnL	1.08%
CAGR	3.88%
Max Drawdown	-16.20%
Annualized Volatility	14.05%
Sharpe Ratio	0.13
Max Favorable Excursion (MFE)	12.54%
Max Adverse Excursion (MAE)	-34.33%
Time Underwater	60.66%
Tail-Risk Drawdown (Worst 5%)	-41.76%

Key Professional Observations

1. Negative Skew — The Central Risk

The strategy's most critical characteristic is its negative skew: average win (+1.75%) is smaller than average loss (-3.00%), yielding a Win/Loss Ratio of only 0.58. This is the inverse of trend-following and is the defining feature of mean-reversion. A Profit Factor of 1.64 confirms the strategy has positive mathematical expectancy — but it requires the high win rate (73.8%) to sustain it. Any deterioration in win rate risks turning the edge negative.

2. The "Dead Zone" Insight

MAE analysis reveals that trades suffering more than -5% adverse excursion almost never recover to profitability. This suggests a hard stop-loss at -5% to -6% could eliminate tail losses (the -21.67% worst trade) without sacrificing winners. Implementing this stop would likely improve both the Sharpe Ratio and the Profit Factor.

3. System Robustness

With 332 trades over 25 years (~13 per year), the sample size provides extremely high statistical confidence. The Monte Carlo median outcome ($269,743) closely validates the actual backtest result ($268,354), confirming the performance is not the product of lucky trade sequencing.

4. The Psychological Framework — Different from Trend-Following

The strategy is "underwater" (below its previous equity high) for 60.66% of its duration. While this is better than trend-following (SMA200: 88.66% underwater), it still requires discipline. The key psychological challenge is different: rather than enduring long periods of doing nothing, the RSI(3) trader must cope with frequent small losses interspersed with winning streaks.

The Statistical Engine: Win Rate Over Trade Size

The most striking metric is the Win Rate of 73.80%. For a systematic strategy, winning three out of every four trades is psychologically manageable — most retail traders can sustain discipline with this win rate. However, a professional recognises that a Profit Factor of 1.64 is "good but not great." Every dollar lost is compensated by only $1.64 in gains. The margin for error is narrower than a high-Profit-Factor trend system.

> *"In essence: the strategy pays for its high win rate through occasionally absorbing large losses. The mathematics work over 332 trades — but the discipline required is to never exit early from a winner and never hold a loser hoping for RSI recovery."*

Risk Management: The Structural Profile

A professional classifies this system as "Tactical Mean-Reversion."

Comparison	RSI(3) Strategy	S&P 500 Benchmark
Max Drawdown	-16.20%	>-50% (historical collapses)
Annualized Volatility	14.05%	~16% (in-market comparison)
Tail-Risk Drawdown (Worst 5%)	-41.76%	N/A

The Monte Carlo simulation warns of a -41.76% tail-risk drawdown in the worst 5% of cases. An institutional trader would capitalise the account based on this survival threshold, not the optimistic -16.20% historical figure.

Operational Realities and Enhancements

The main professional critique of this strategy is the Sharpe Ratio of 0.13, driven by the negative skew. Three potential enhancements stand out:

- Hard stop-loss at -5% MAE: Would eliminate tail losses and improve Profit Factor
- Position sizing by Kelly: At 28.93% full Kelly, a fractional Kelly of 10–15% would reduce variance while maintaining positive expectancy
- Idle cash deployment: The 58% of time out of the market could earn T-Bill yields, improving the risk-adjusted return profile

Final Verdict

> *"This strategy is effectively a 'Market Insurance Collector' — it consistently earns a small premium (73.8% win rate) by selling the market insurance against short-term oversold conditions, but occasionally must pay out large claims (the -21% worst trade). The mathematics are sound; the discipline required is to never deviate from the system."*

It is designed for a practitioner who prioritises capital preservation over aggressive alpha, values statistical robustness (332 trades), and can mechanically execute entries and exits without discretionary override. The strategy's low maximum drawdown (-16.20%) and near-zero probability of ruin make it a structurally sound core holding in a diversified systematic portfolio.

Classification	Analysis Period / Asset Class
Tactical Mean-Reversion Strategy Analysis	2000–2025 | S&P 500